Introduction

Instructor: Xuanji Yu

Courtesy of Professor Marie Roch

Chapters 1 & 2, 27.3, Russell & Norvig

What is AI?

"AI in Science Fiction"

HAL-9000
2001: A SPACE ODYSSEY (1968)

What is AI?

Riken and Sumitomo nursing robot

“Machines with human skills”

- Michelle Zhou, CEO Juji Inc.

Intelligence per Merriam Webster

1 a : 1) the ability to learn or understand or to deal with new or trying situations **:** REASON; *also* **:** the skilled use of reason

2) the ability to apply knowledge to manipulate one's environment or to think abstractly as measured by objective criteria (as tests)

b *Christian Science* **:** the basic eternal quality of divine Mind

c : mental acuteness **:** SHREWDNESS

2 a : an intelligent entity; *especially* **:** ANGEL

b : intelligent minds or mind <cosmic *intelligence*>

3 : the act of understanding **:** COMPREHENSION

Artificial intelligence

- weak AI - Machines act as if they are intelligent

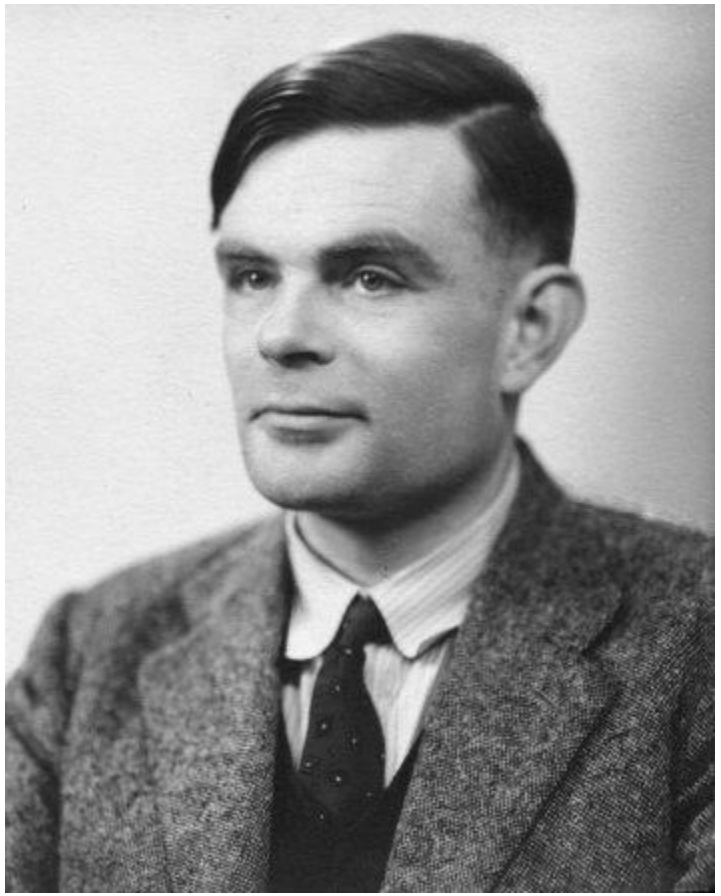
- strong AI – Machines think

Brent Spiner as Commander Data
Star Trek Next Generation

What does it mean for a machine to be intelligent?

Turing Test – Can a human tell that they are interacting with a computer?

Variant of this competition occurs today with Loebner Prize (very restricted Turing test for chatbots)



Alan Turing 1912-1954
See *The Imitation Game* (2014) for a dramatization of his life.

Eliza (Weizenbaum, 1966) - Early natural language processing computer program

The most famous script, DOCTOR, simulated a psychotherapist of the Rogerian school (in which the therapist often reflects back the patient's words to the patient)

young woman: Men are all alike.

eliza: In what way?

young woman: They're always bugging us about something specific or other.

eliza: Can you think of a specific example?

young woman: Well, my boyfriend made me come here.

eliza: Your boyfriend made you come here?

young woman: He says I'm depressed much of the time.

eliza: I am sorry to hear you are depressed.

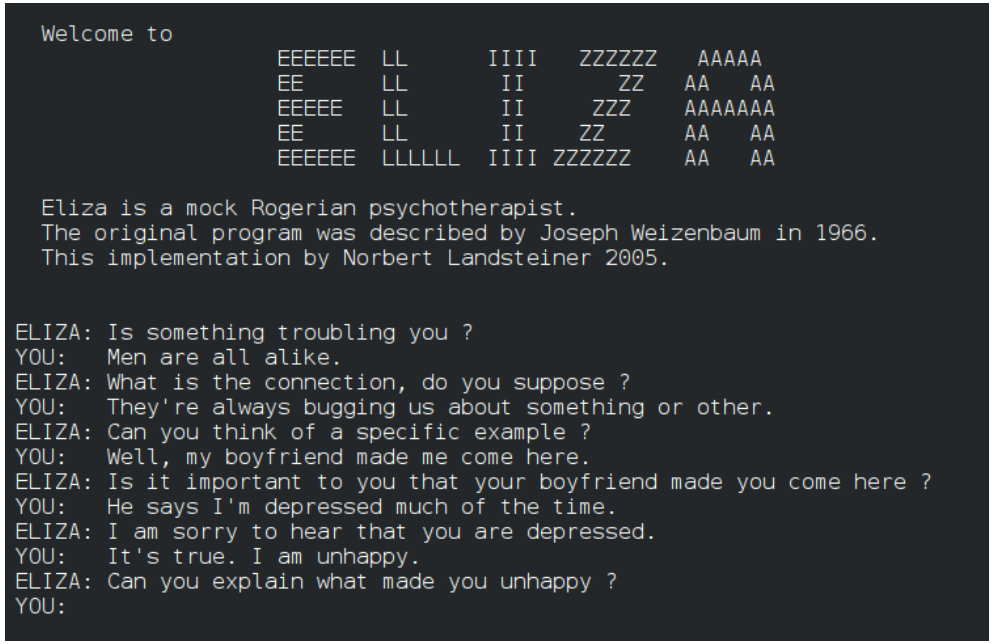
young woman: It's true. I'm unhappy.

eliza: Do you think coming here will help you not to be unhappy?

young woman: I need some help, that much seems certain.

eliza: What would it mean to you if you got some help?

young woman: Perhaps I could learn to get along with my mother.



Simple rules ---> disappointment

young woman: Men are all alike.
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(X me Y) ---> (X you Y)

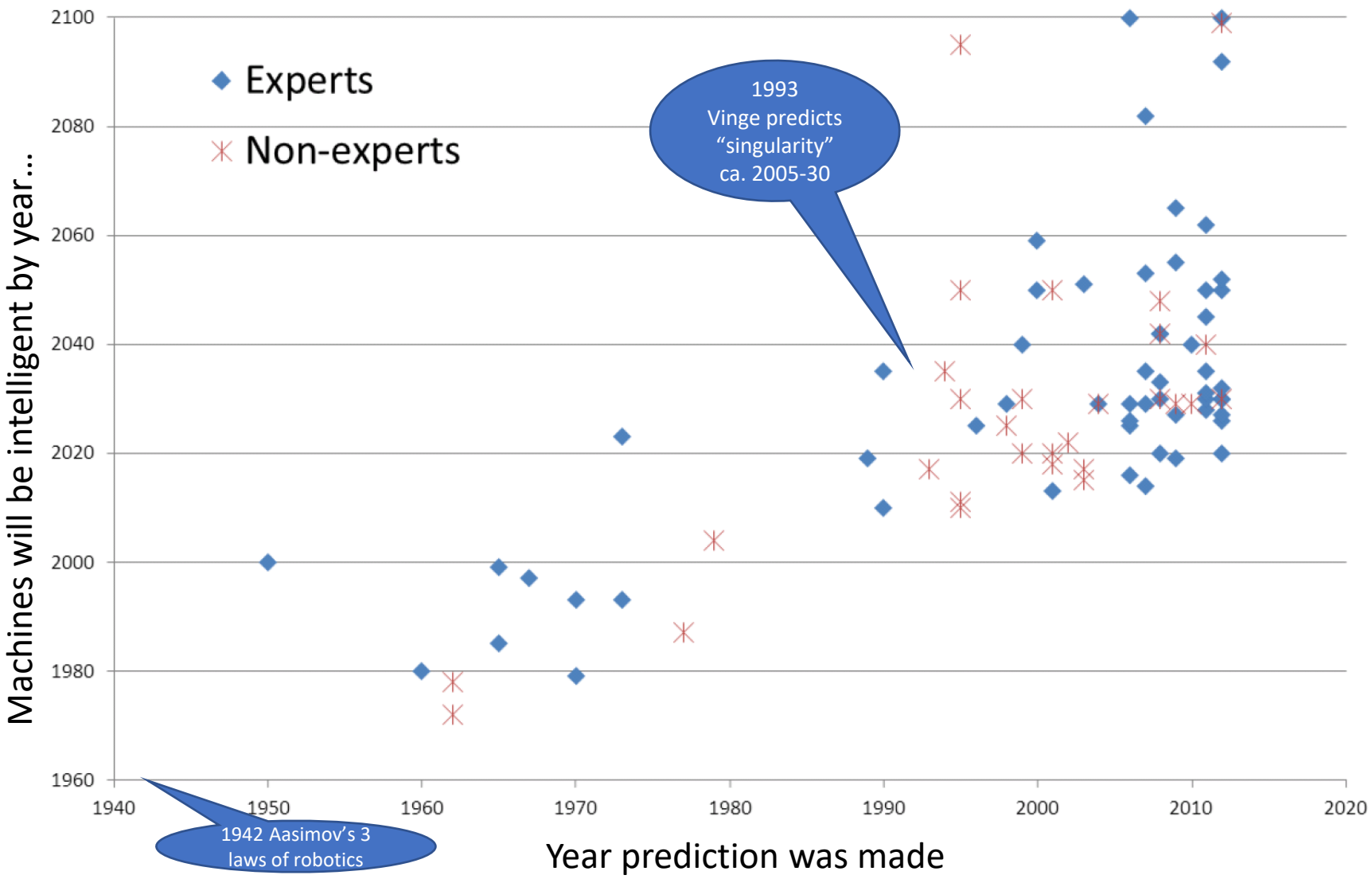
(I remember X) ---> (Why do you remember X just now?)

(My {family-member} is Y) ---> (Who else in your family is Y?)

(X {family-member} Y) ---> (Tell me more about your family)

keywords with ranked rule

When will we have strong AI? (Armstrong, et al. 2014)



“Singularity" refers to a hypothetical future point in time when technological growth becomes uncontrollable and irreversible, leading to unforeseeable changes in human civilization.

Areas contributing to AI

- Philosophy
- Mathematics
- Neuroscience
- Psychology
- Linguistics
- Computer science

- Many application areas contribute as well (e.g. economics)

OpenAI Defines Five Stages To Track Progress Toward Human-Level Intelligence

Stages of Artificial Intelligence	
Level 1	Chatbots, AI with conversational language
Level 2	Reasoners, human-level problem solving
Level 3	Agents, systems that can take actions
Level 4	Innovators, AI that can aid in invention
Level 5	Organizations, AI that can do the work of an organization

Source: Bloomberg reporting

Bloomberg

Different Approaches to Define AI

Approach	Focus	AI Goal
Turing Test	Acting humanly	Behavior indistinguishable from human
Cognitive Approach	Thinking humanly	Mimic human mental processes
Laws of Thought	Thinking rationally	Use formal rules of logic
Rational Agent	Acting rationally	Make best decisions with available info

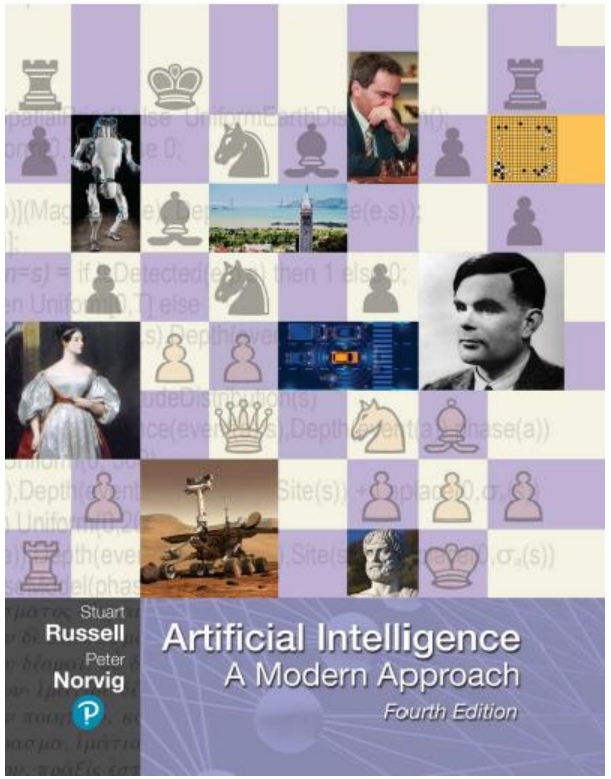
CS 450 Definition of ‘AI’

- This class focuses on the ‘action’ aspect of AI or ‘usefulness’ of AI
  - don’t care about how AI think

AI is science of making machine

Thinking Humanly	**Thinking Rationally**
“The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense.” (Haugeland, 1985) “[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . .” (Bellman, 1978)	“The study of mental faculties through the use of computational models.” (Charniak and McDermott, 1985) “The study of the computations that make it possible to perceive, reason, and act.” (Winston, 1992)
Acting Humanly	**Acting Rationally**
“The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990) “The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)	“Computational Intelligence is the study of the design of intelligent agents.” (Poole *et al.*, 1998) “AI . . . is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)

Figure 1.1 Some definitions of artificial intelligence, organized into four categories.



CS450 AI: Design Agent Acting Rationally

The concept of AI "acting rationally" revolves around making decisions that maximize the achievement of pre-defined goals. Key ideas:

- Rationality and Goals:
  - Being rational in AI means making decisions that effectively achieve the goals that have been set.
- Focus on Decisions, Not Thought Process:
  - The emphasis is on what actions are taken and whether they lead to the desired outcomes.

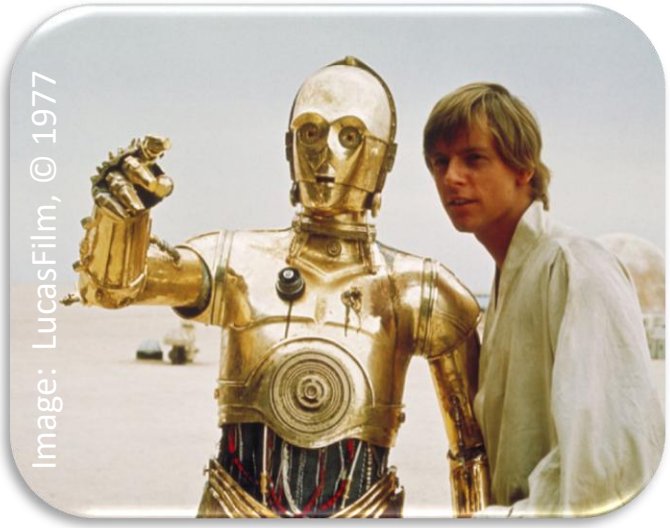
CS450 AI: Design Agent Acting Rationally

The concept of AI "acting rationally" revolves around making decisions that maximize the achievement of pre-defined goals. Key ideas:

- Utility of Outcomes:
  - Utility refers to the measure of how desirable or valuable a particular outcome is relative to achieving the goal.
- Maximizing Expected Utility:
  - Rational behavior, therefore, involves maximizing the expected utility of actions.

AI Agent Acting Rationally

Intelligent agents



- Sensors provide perceptual input (**percepts**) of environment

- Agent makes decisions

- Actions carried out through actuators that may affect the environment

Intelligent agents

- Softbot – Software only agent
  - Available data are percepts
  - Examples: web-based reputation monitoring, game opponent

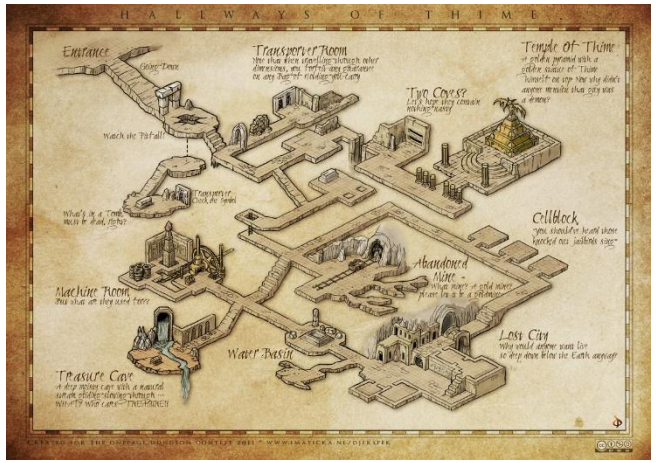
Task environments

In what environment will the agent be operating?

- fully vs. partially observable

  partially observable → uncertain state

- Rules are
  - known: Outcome (or outcome probabilities) are known
  - unknown – Outcomes must be learned



Task environments

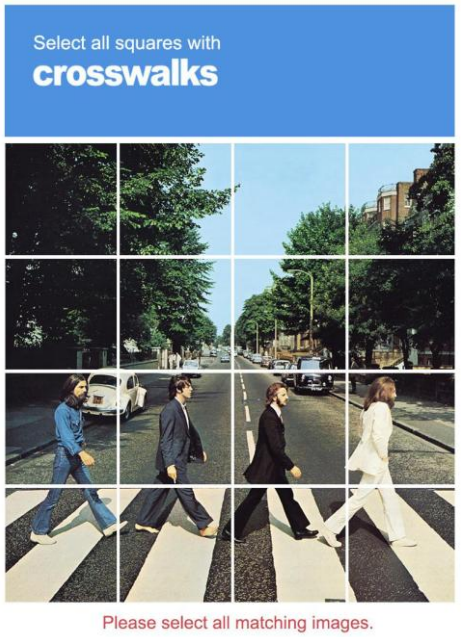
Getty images

NASA Perseverance

- single- vs. multi- agent
- multiagent issues
  - cooperative vs. competitive
  - communication
  - randomization to prevent predictability

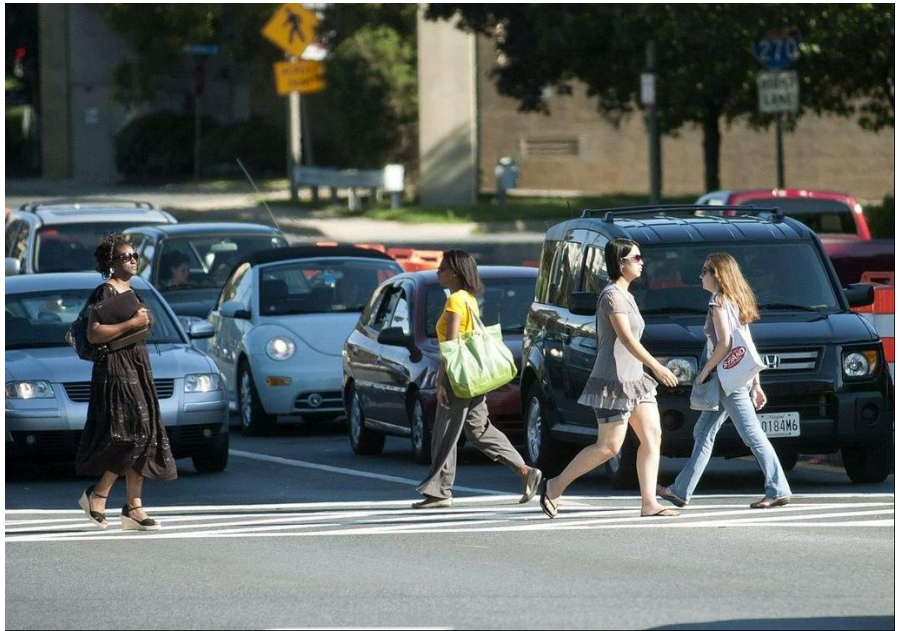
Task environments

- What happens when an agent acts?
  - deterministic – we know next state
  - stochastic
    - nondeterministic factors may influence (stochastic →probabilities) leading to an *uncertain* state
- Decisions are
  - episodic – Next decision only depends on state
  - sequential – Next decision dependent on previous ones and can affect future states



Task environments

- State can be
  - static – does not change while agent is deciding next action
  - dynamic – Environment constantly changing
  - semidynamic – Environment static, but performance is time dependent



See Figure 2.6 (p. 47) for example task environments

Agent structure

- An agent's architecture consists of
  - data structures
  - code
- Simplest agent: table driven
  - relies on a precomputed table that maps inputs or states to specific actions

```
function table-driven-agent(percept) returns action
  persistent: percepts (sequence, empty at first)
              table of actions indexed by
                     percept sequence


  percepts.append(percept)
  return lookup(percepts, table)
```

What's wrong here?

Agent structure

Simplest agent - table driven agent can be suitable for certain well-defined and static environments where adaptability and learning are not critical requirements, but they have the following limitations:

- Memory Usage
- Limited Adaptability
- Maintenance Challenges
- Scalability Issues
- Hardcoding Limitations
- Not Suitable for Continuous Domains
- Lack of Generalization
- Inefficient for Large Databases
- Dependency on Domain Expertise

Agent types

- Simple reflex – Reacts to stimulus
  - relies solely on a set of condition-action rules
- Model-based reflex – Stimulus + state
  - maintains an internal representation or model of the world, estimating the current state of the environment
- Goal-based – Work towards objective
- Utility-based – Increase value of something measurable
  - makes decisions by considering the expected utility of different actions in achieving its goals
- Learning – Adjust goals/utility/rules over time

Simple-reflex agents

- Ignores percept history, uses the current one
- Productions (aka conditions-action) decide action, e.g.

  person waving → wave

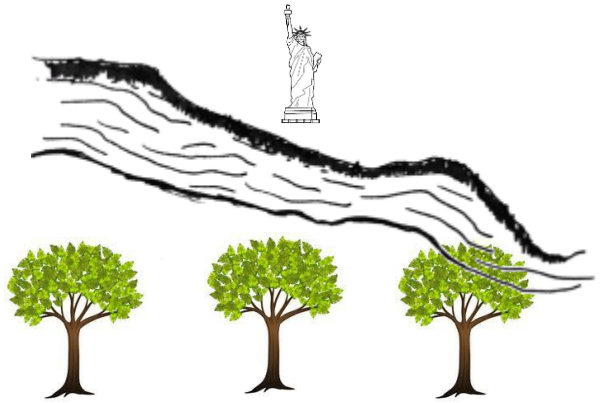
  person smiling → smile

  person swinging hammer towards me → duck!

source: Imperial College London

Model-based reflex agents

- Add internal state
  - maintains an internal representation or model of the world, estimating the current state of the environment
- New percepts update the state
- Productions based on percept and state



no credit available

The Nature Conservancy

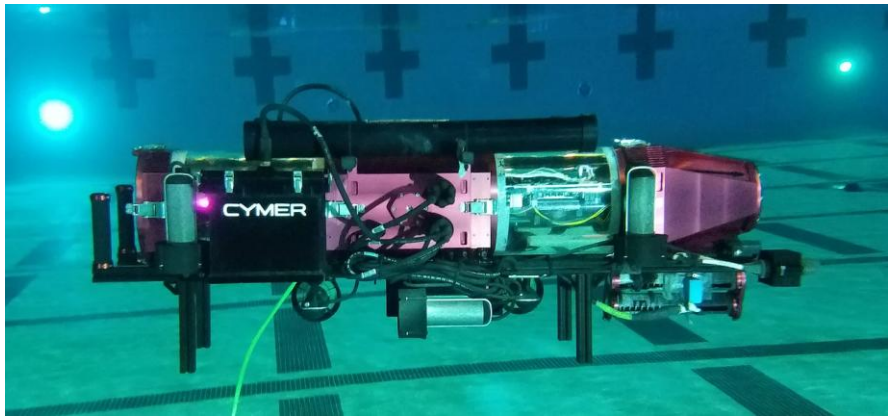
States in model-based agents

States can be

- atomic – state is indivisible, it does not have multiple parts
  sometimes we treat things as atomic, even if they are not: e.g. configuration of pieces on a chess board

- factored – multiple attributes
  example: autonomous underwater vehicles have representations for pressure, power, buoyancy, sonar, vision, …

SAN DIEGO STATE UNIVERSITY
MECHATRONICS



SDSU Mechatronics Robosub AUV

Goal-based agents

- Agent works to achieve a specific state
- Usually requires: Search and Planning

Utility-based

- Based on utility theory: The idea of how useful or happy something makes you.

omelet.us

15 utils

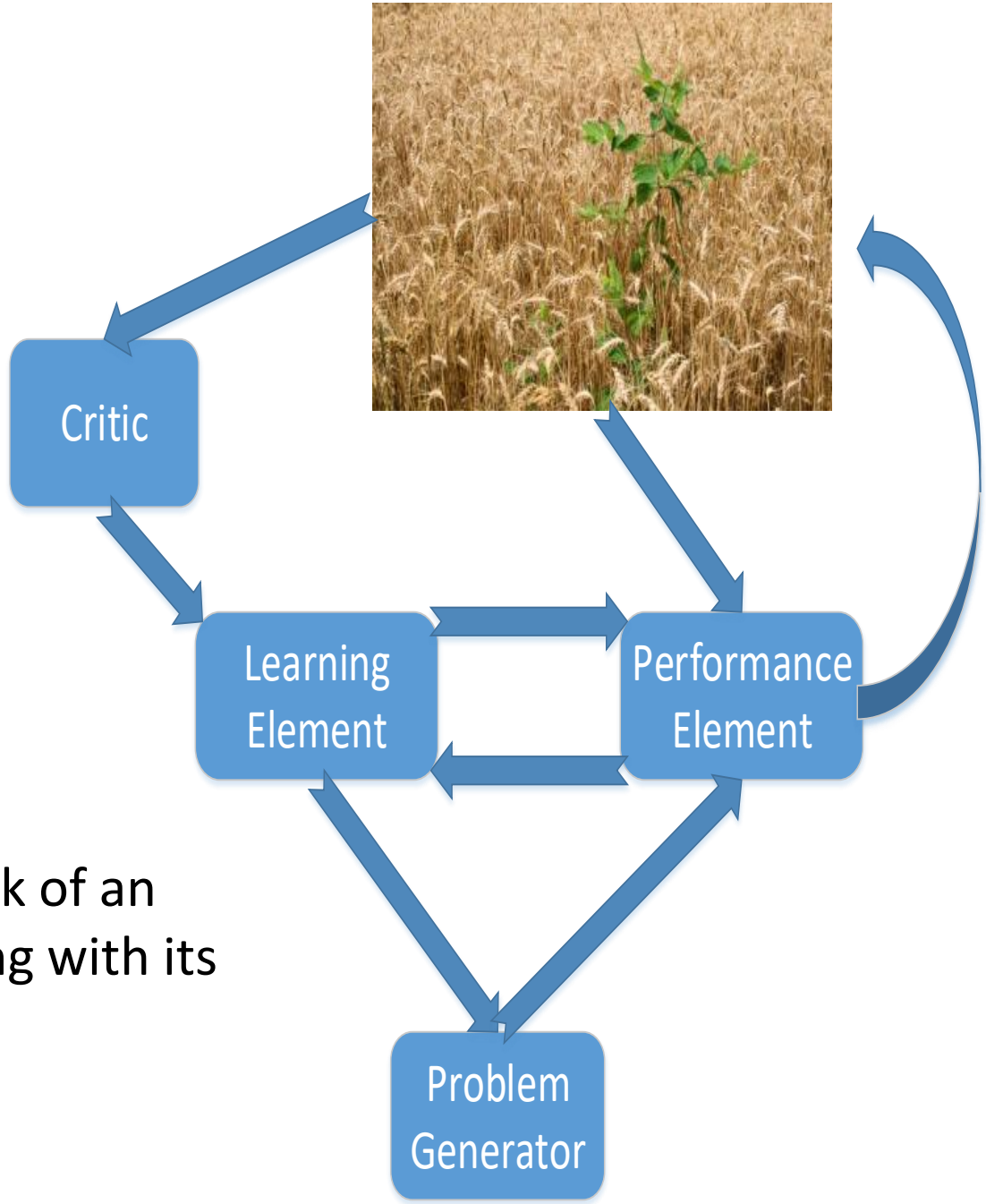
hellotushy.com

-5 utils

- Decisions are made to maximize the expected utility.

Learning agents

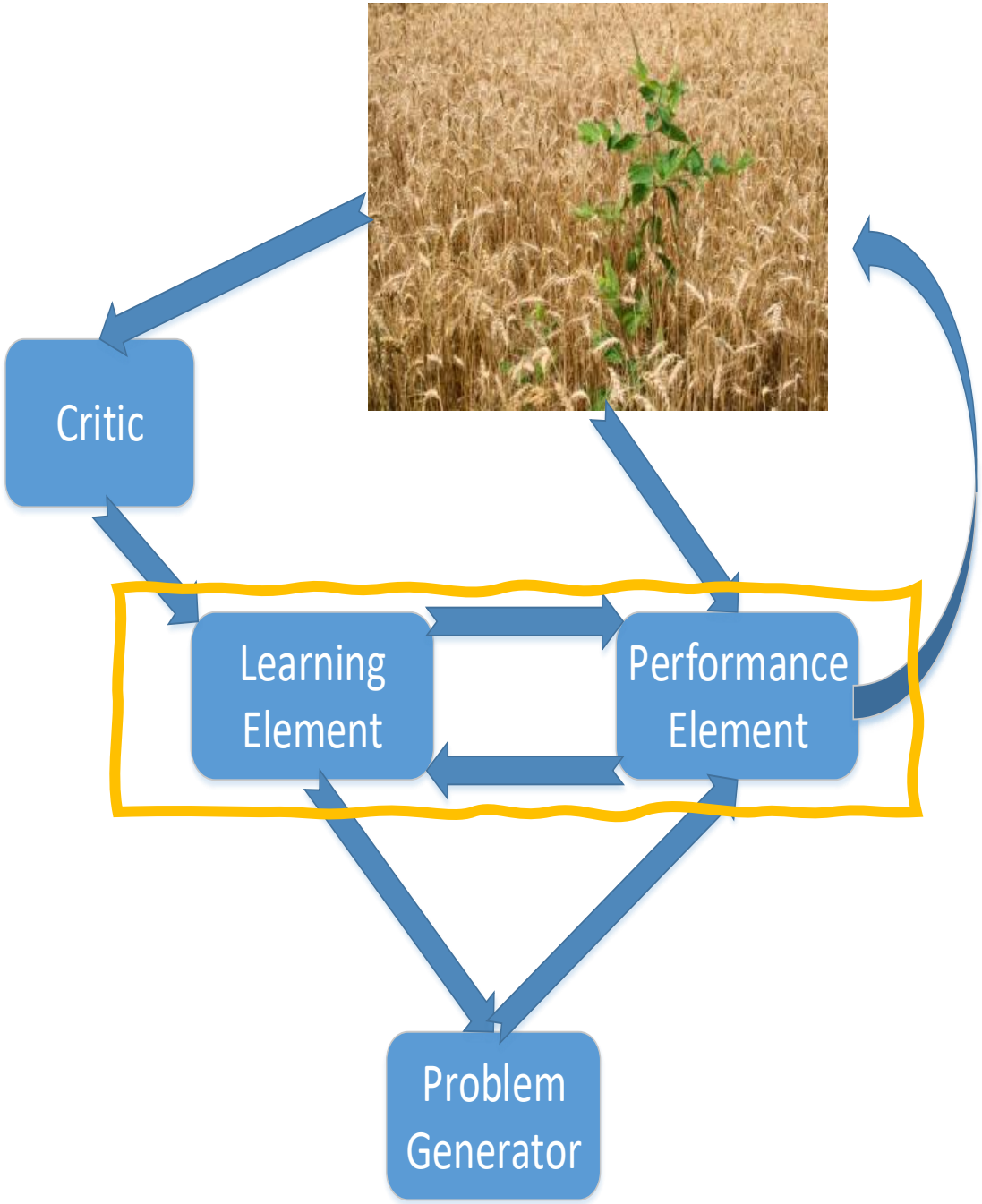
Decision rules are adjusted based on performance over time



Basic framework of an agent interacting with its environment

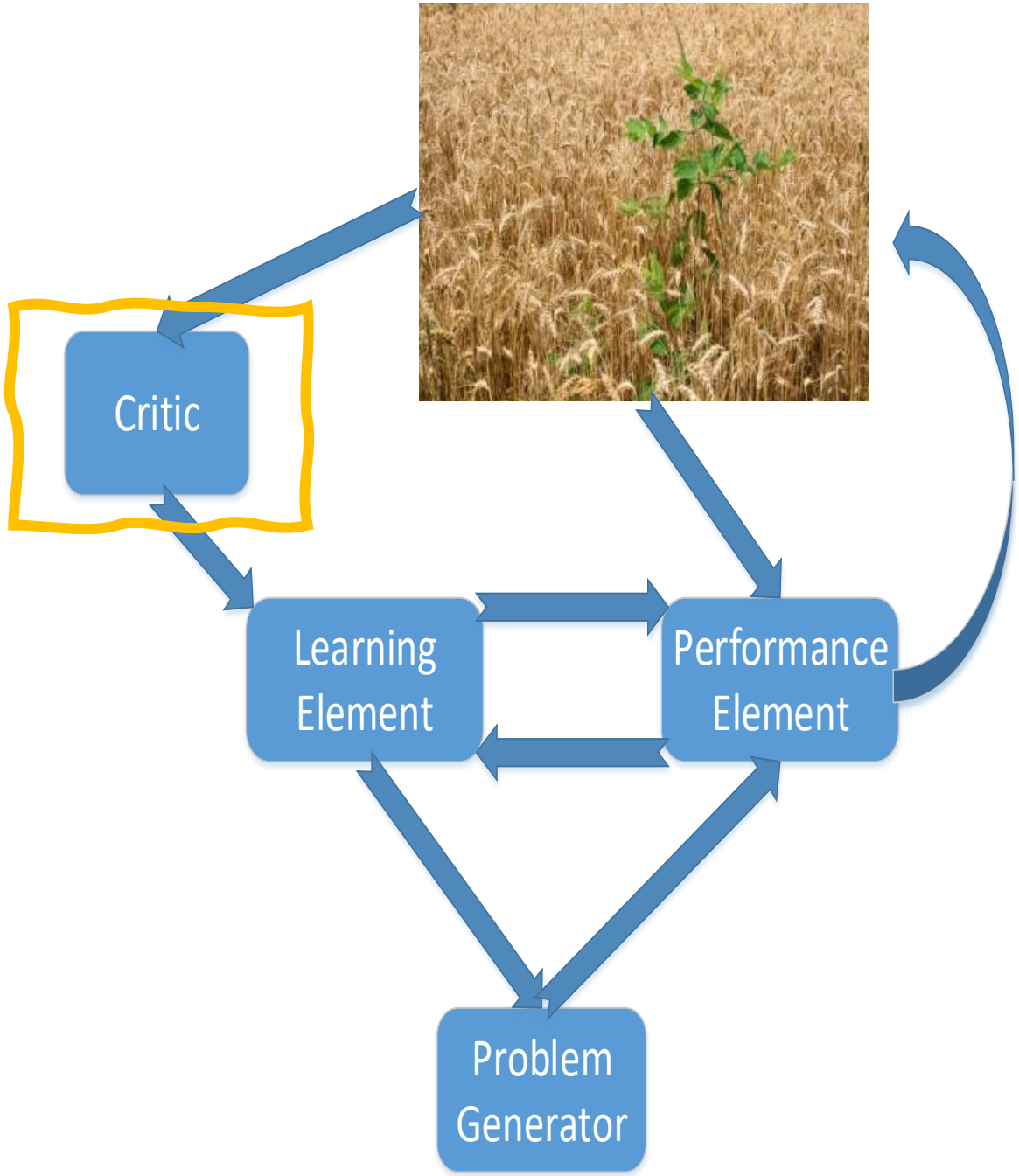
Learning agents

- Performance element makes decisions based on percepts
- Learning element uses knowledge from performance element to modify rules



Learning agents

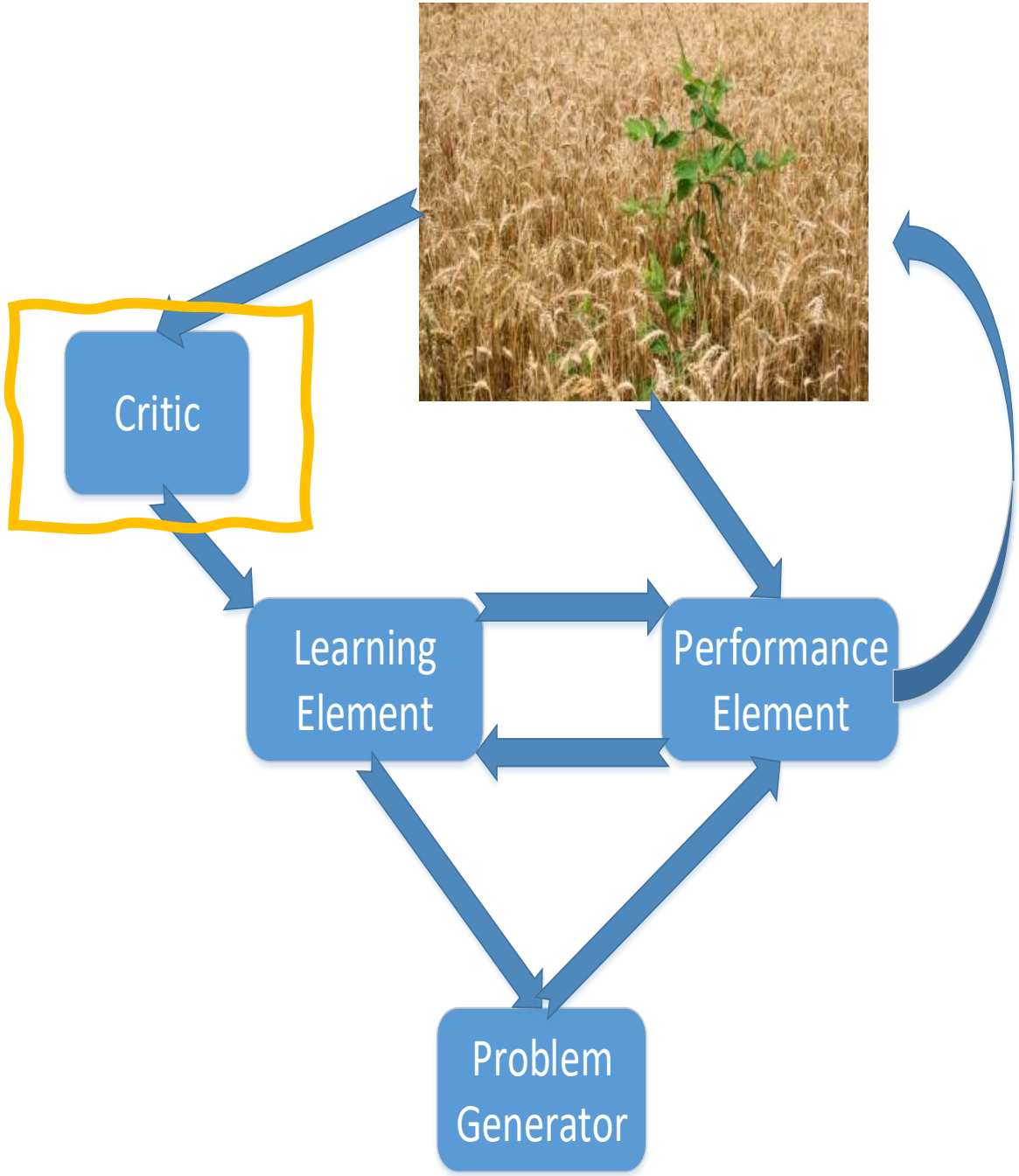
Critic monitors the environment to see how the system is doing and keeps learning element from going off track.



Learning agents

Critic monitors the environment:

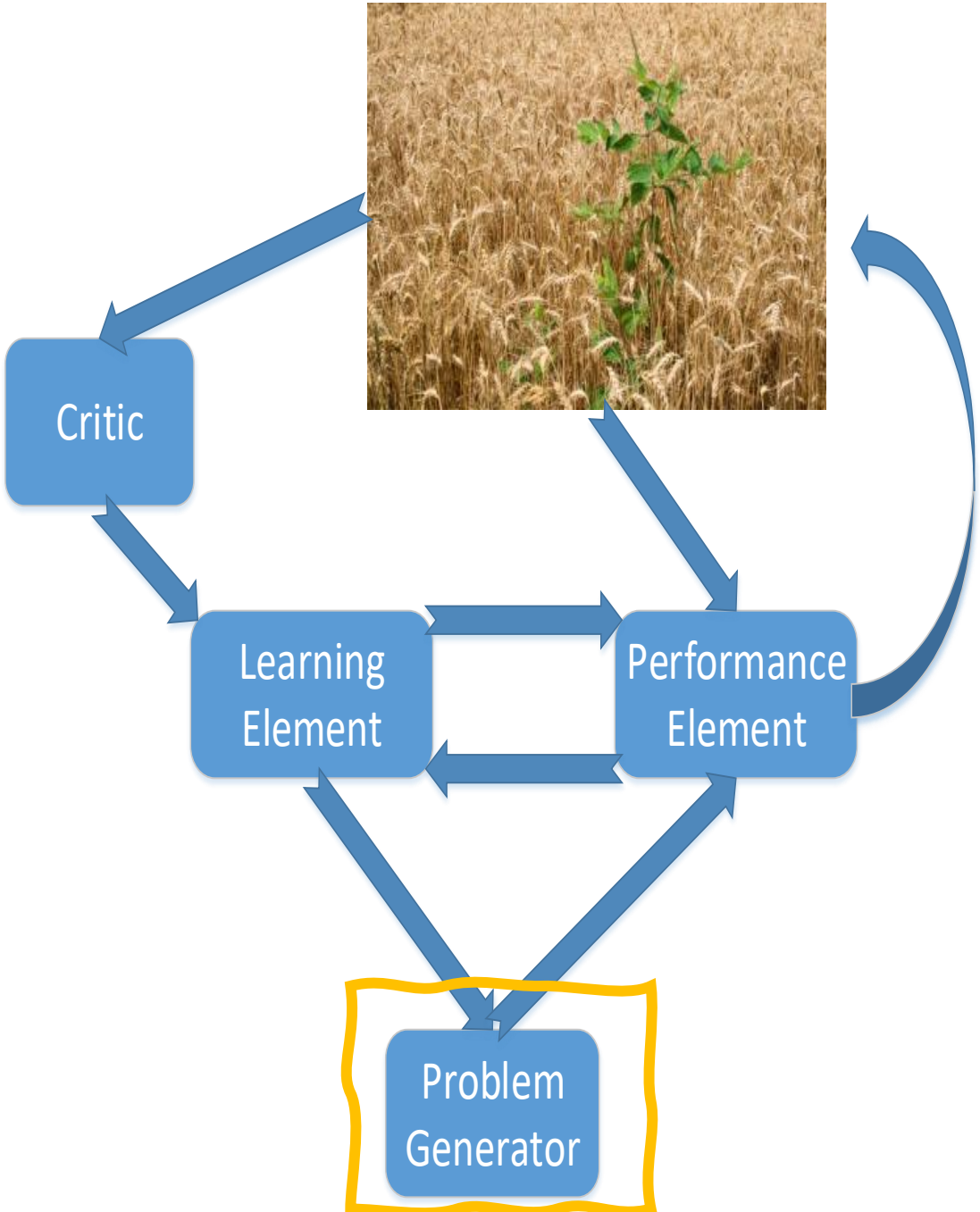
- Provides utility of the current percepts according to an unchanging standard
- Allows learning element to learn useful goals



Learning agents

Problem generator

- Uses learning element's goals to suggest experiments or actions or scenarios that challenge the agent and contribute to its learning process.
- Experiments may lead to the learning element improving the performance element

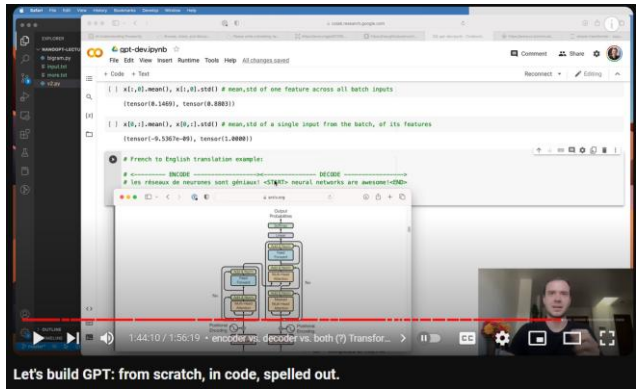


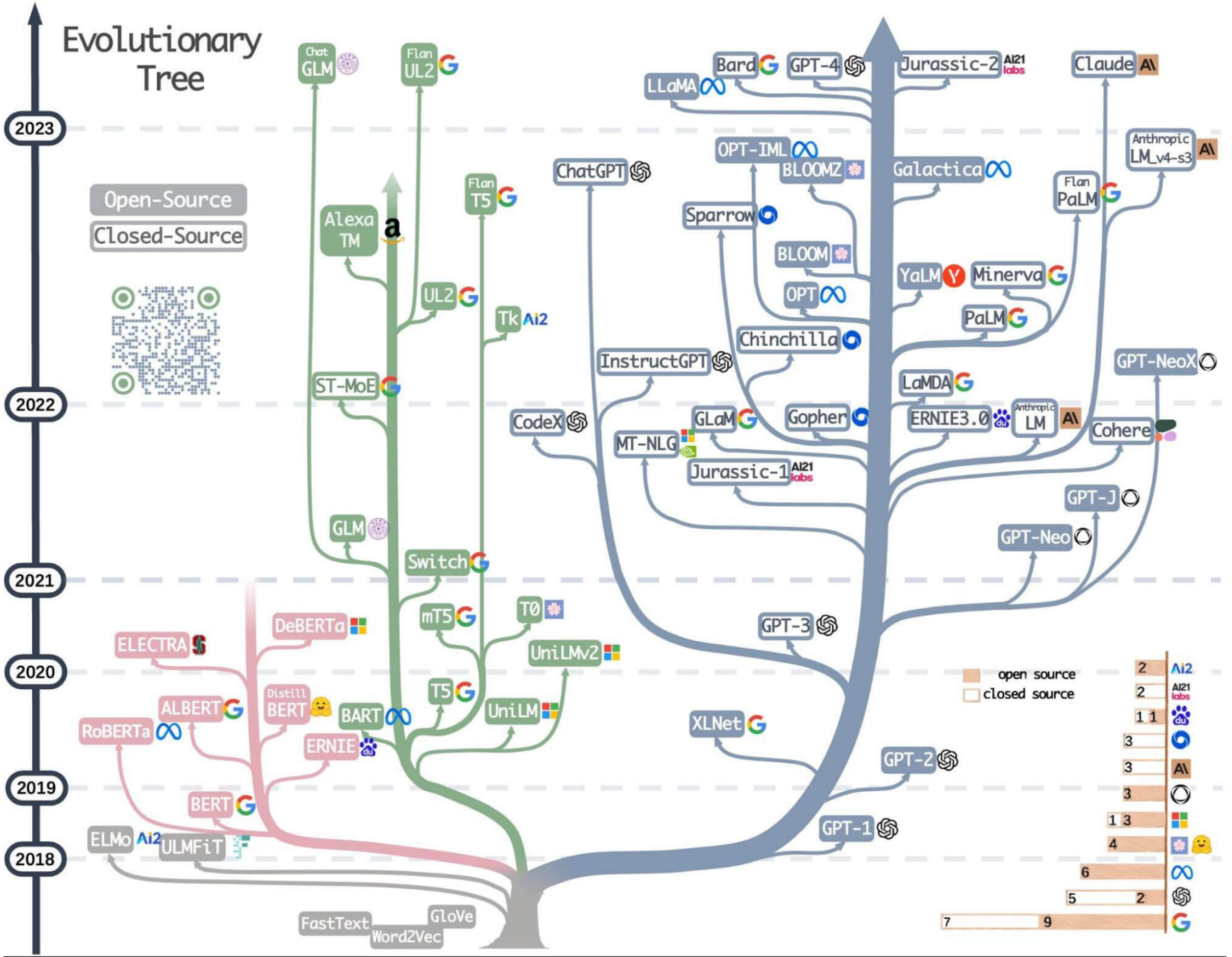
LLM

Deep Learning

- Text->Token
- Predict next Token
- Transformer

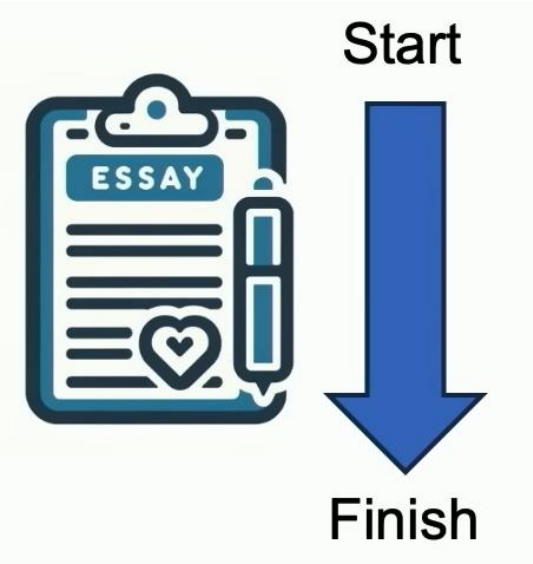
- Andrej Karpathy
- [Build GPT]



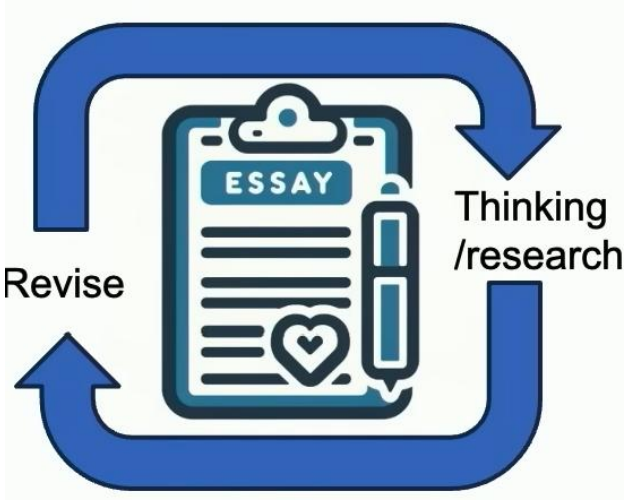


Large Language Model LLM-based agent

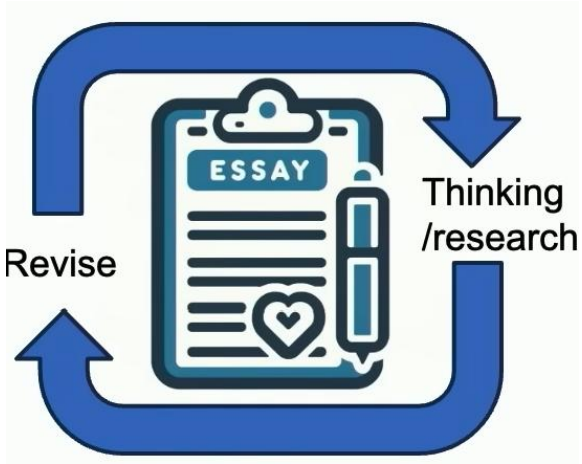
- Non-agentic workflow (zero-shot)



- Agentic workflow
  - Decision-Making
  - Action



LLM-based agent

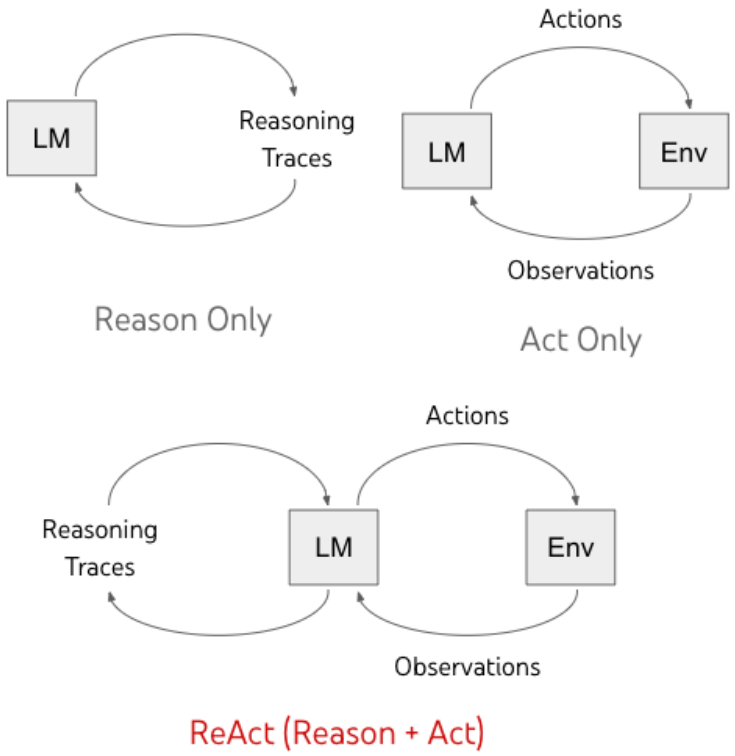


- Agentic Reasoning Design Patterns
  - Reflection
    - Self-reflection, e.g., write code
      - and check for correctness, style, and efficiency
  - Tool use
    - Analysis
      - e.g, code execution
    - Research
      - e.g., search engine
    - Productivity
      - e.g, email
    - Images
      - e.g. Dall-E

[OpenAI Agent and Operators]

LLM-based agent

- Agentic Reasoning Design Patterns
  - Planning
    - Chain of thought
    - Reason + Act (ReAct)



HotPotQA

Question: Seven Brief Lessons on Physics was written by an Italian physicist that has worked in France since what year?

(a) Standard

Answer: 1986 ✗

(b) Reason only

Thought: Let's think step by step. Seven Brief Lessons on Physics was written by Carlo Rovelli. Carlo Rovelli has worked in France since 1990.

Answer: 1990 ✗

(c) Act only

Act 1: Search[Seven Brief Lessons on Physics]
Obs 1: Seven Brief Lessons on Physics (Italian: Sette brevi lezioni di fisica) is a short book by the Italian physicist Carlo Rovelli. Originally published in Italian in…

Act 2: Lookup[Carlo Rovelli]
Obs 2: (Result 1 / 1) Seven Brief Lessons on Physics (Italian: Sette brevi lezioni di fisica) is a short book by the Italian physicist Carlo Rovelli.

Act 3: Finish[1983] ✗

(d) ReAct

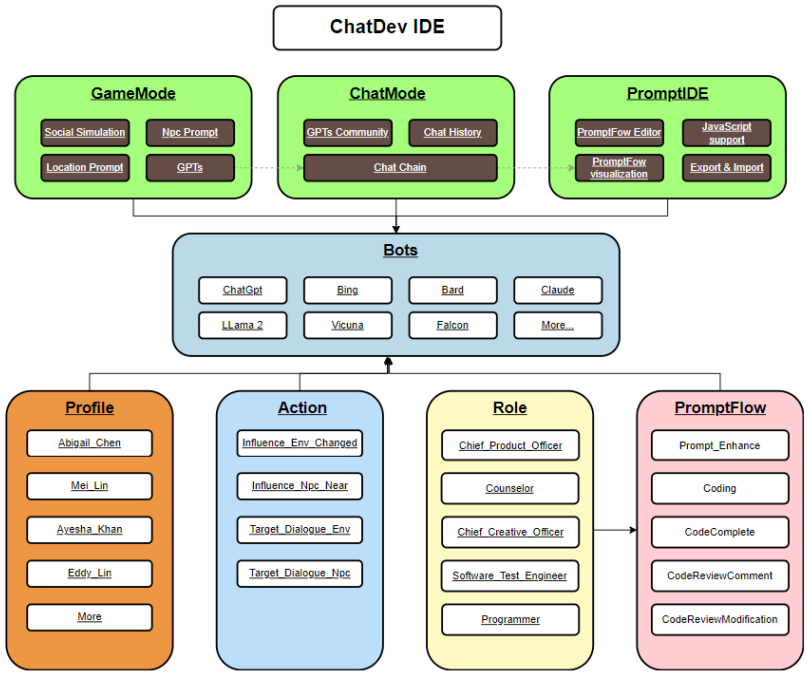
Thought 1: I need to search Seven Brief Lessons on Physics, find its author, then find when the author has worked in France since.
Act 1: Search[Seven Brief Lessons on Physics]
Obs 1: Seven Brief Lessons on Physics (Italian: Sette brevi lezioni di fisica) is a short book by the Italian physicist Carlo Rovelli. Originally published in Italian in…

Thought 2: The author of Seven Brief Lessons on Physics is Carlo Rovelli. I need to search Carlo Rovelli next and find when he has worked in France since.
Act 2: Search[Carlo Rovelli]
Obs 2: Carlo Rovelli is an Italian theoretical physicist and writer who has worked in Italy, the United States and, since 2000, in France.[1] He is also currently a Distinguished Visiting Research Chair at the Perimeter Institute…

Thought 3: Carlo Rovelli has worked in France since 2000. So the answer is 2000.
Act 3: Finish[2000] ✓

LLM-based agent

- Agentic Reasoning Design Patterns
  - Multiagent collaboration
    - ChatDev AI Town Social Simulation
    - AutoGen (Microsoft)
    - CrewAI (OpenAI)



AI Issues

Societal issues

Job displacement

- Which job is more likely to be replaced by AI
  - White collar
    - with art major
    - with STEM major
  - Blue collar

Societal issues

- White collar with art major: Natural language response
  - Personal assistants
  - Call centers
  - Document processing
  - Games and entertainment
  - Education

Societal issues

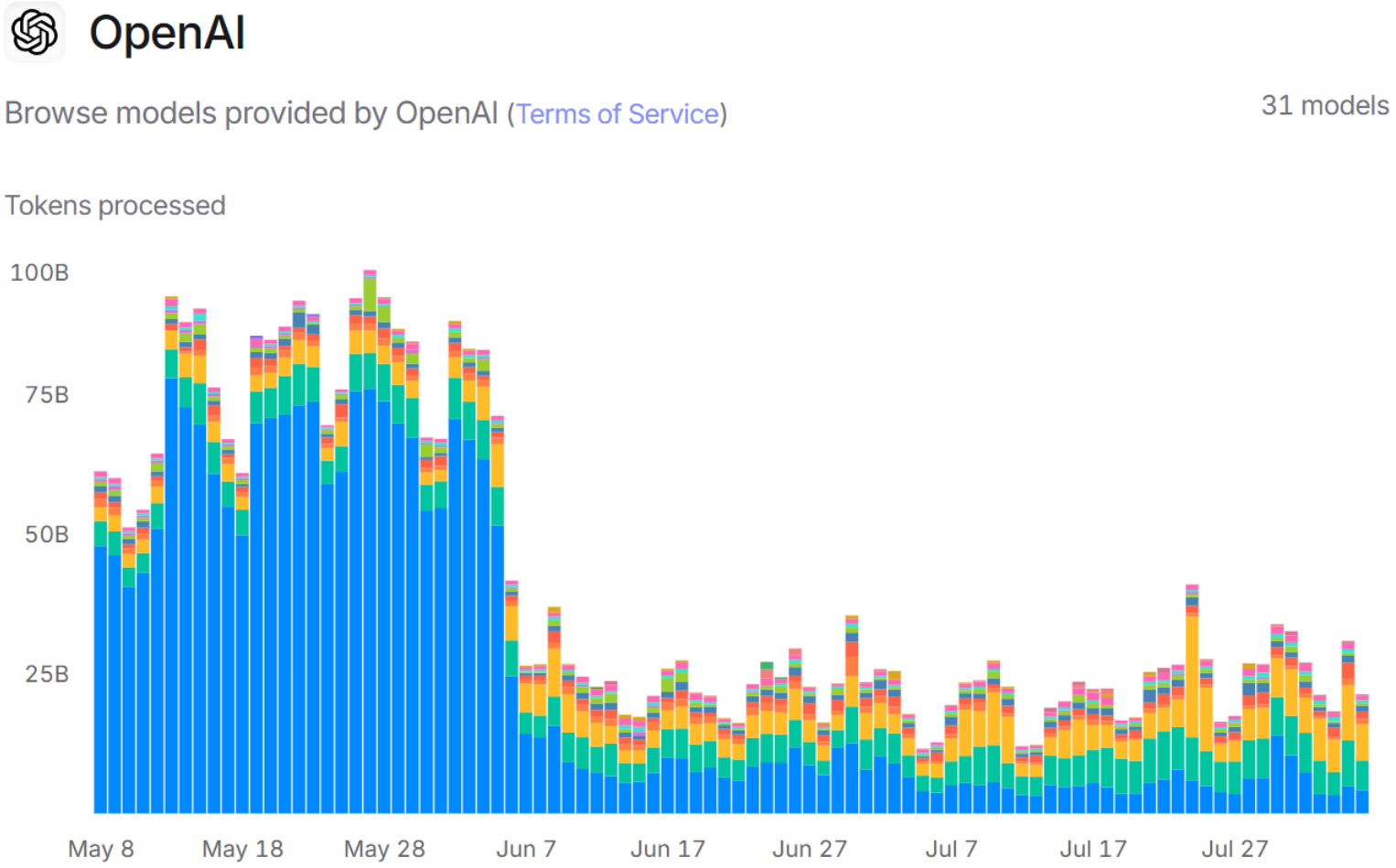
- White collar with STEM major: output code, problem solution
  - Software engineering

- Blue collars: understand world, output action
  - Self-driving
  - Robotics

Societal issues

Impact on education

- This class you're welcome to use AI tools, like ChatGPT (SDSU ChatGPT Edu), Claude AI, Microsoft Copilot, or Gemini (Free Pro Plan for Students), to help with assignments and projects, as long as you write everything in your own words and understand what you submit.
- Final – Closed book



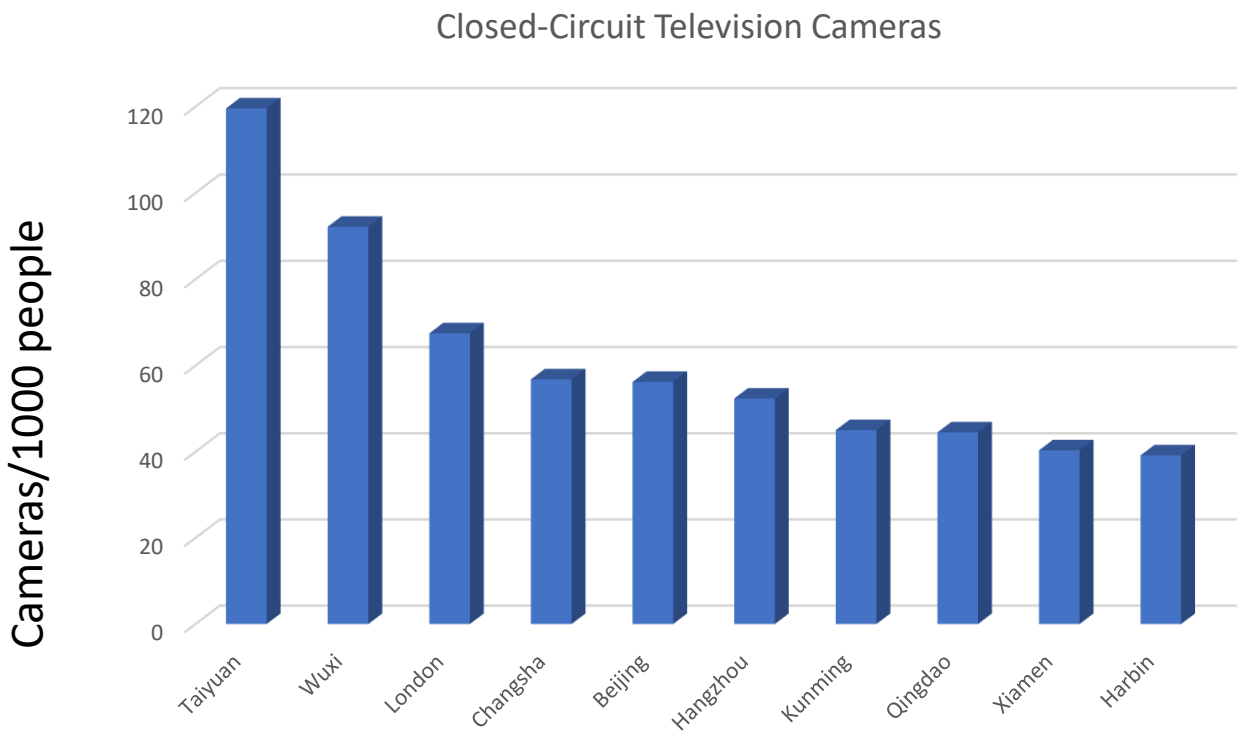
Ethical issues

- Autonomous weapon systems
  - Is it ethical to have lethal systems without a human in the loop?
  - Issues:
    - Should machines be making decisions that result in loss of life?
    - What about errors?
  - Example: Harop missile – Can be configured to autonomously recognize targets and launch.

wikipedia.org/wiki/IAI_Harop

Ethical issues

- Surveillance, security, privacy



Data Source:
Keegan, Matthew, "The Most Surveilled Cities in the World", *US News and World Report*, August 14, 2020.

Ethical issues

- Surveillance – Cameras, microphones, online activity can be coupled with AI systems to recognize, track, and flag people/behaviors
  - Government and corporate users
  - Misidentification can create large problems for individuals

Thumbnails from Cao et al's Open Pose (2019)
full video

Ethical issues

- Security/Privacy
  - As data are amassed, misuse can occur from
    - corporate/government actions
    - data breaches
  - Protection strategies
    - De-identification – Remove data that can be linked to individuals
    - Problem: Remaining data can be traced to individuals
      Sweeney showed that 87% of Americans can be uniquely identified from
      - date of birth
      - gender
      - zip code
  - People frequently do not know what data have been amassed and how it is used.

Ethical issues

- Fairness and bias
  - Machine learning algorithms learn from the data they are given
  - Example: face recognition (Manyika, Silberg, Presten (2019) *Harvard Business Review*)
    - Joy Buolamwini found higher errors on minority women
      (and minorities in general).
    - Potential root cause: lack of representation in training data

Joy Buolamwini MIT News

Ethical issues

Being chauffeured by a Tesla
(don't do this)

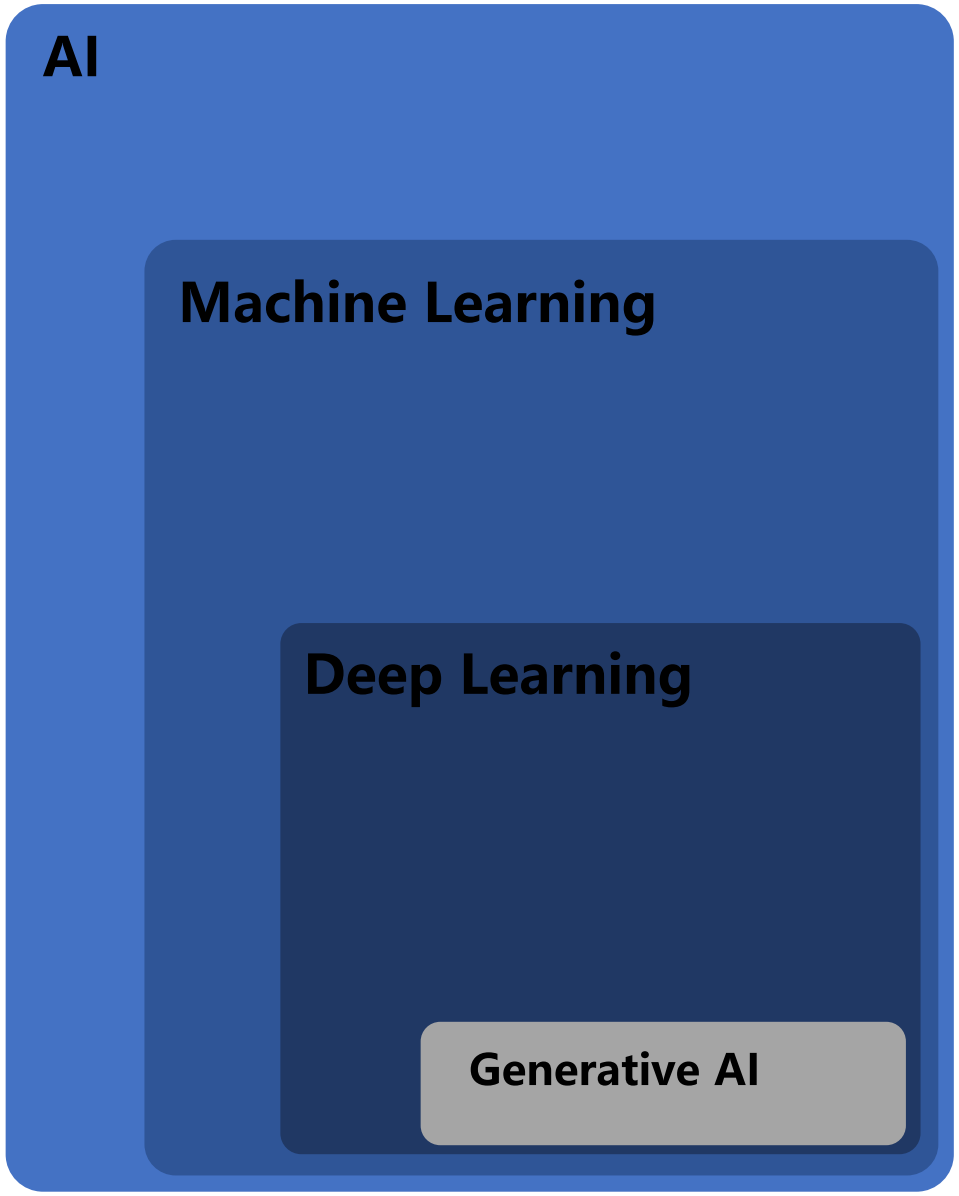
- Trust and transparency
  - How good are our systems?
    - Are they robust?

deepdrive.berkeley.edu: Stop signs recognized as 45 MPH speed limit, right turn as stop

  - What have we done to validate that they work?
  - Are decisions *explainable*?

Other Related Topics

AI vs ML etc. (from Microsoft)



AI

Field of computer science that seeks to create intelligent machines that can replicate or surpass human intelligence

Machine Learning

A subset of artificial intelligence that enables machines to learn from existing data and improve that data to make decisions or predictions

Deep Learning

A machine learning technique that uses multi-layered neural networks to process data and make decisions

Generative AI

creates new written, visual, and auditory content based on prompts or existing data

Model: LLM are still ML models, but 100x Larger

- Model training components
  - Data
  - Algorithm
  - Compute
- Cooking
  - Raw food
  - Recipe
  - Cookware, oven

Stargate AI Infrastructure Project
$500 Billion

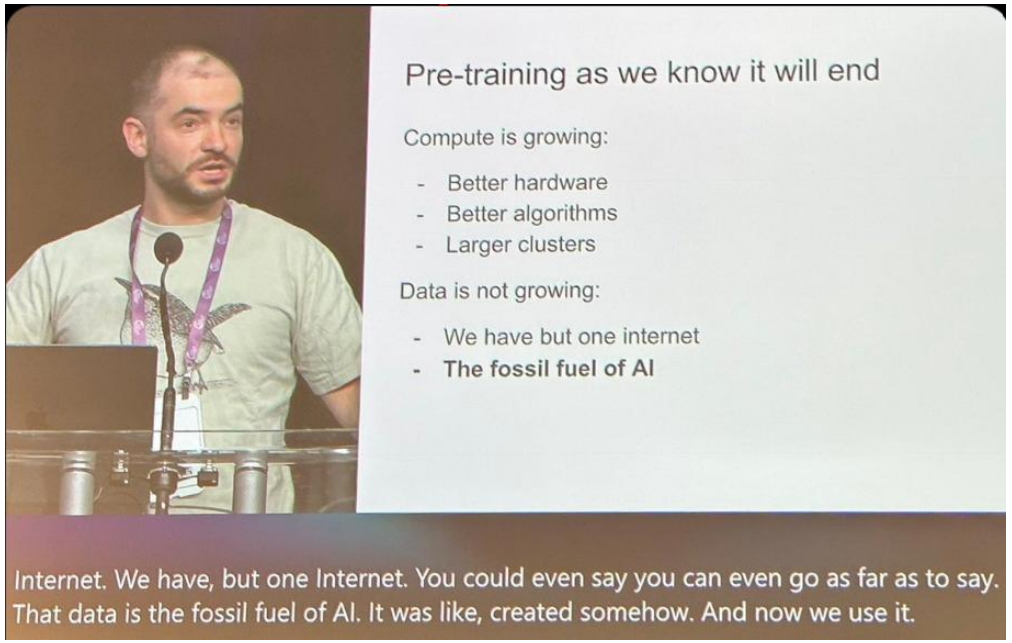
Most lessons learned from previous ML models sill apply to LLM

Data

```python
from datasets import load_dataset, DatasetDict

# Load dataset
raw_datasets = load_dataset("wikipedia", "20220301.en")
```

- (High quality) Data Source
  - Internet (is not growing, Ilya NeurIPS 2024)
  - Synthetic Data?



- Tools
  - For Data Collection: BeautifulSoup, Scrapy.
  - For Cleaning: NLTK, spaCy, langdetect.
  - For Annotation: Label Studio, Prodigy.
  - For Preprocessing: Hugging Face Transformers, FastText.

Compute Hardware

- Bandwidth
  - Distributed, bandwidth bottleneck
- Memory
  - Limit model size
- Compute
- Resources
  - Electricity
  - Price

Nvidia: During a gold rush, sell shovels

- cuda: high-level programming language for graphic card that software can use to spread complex maths like AI into smaller, parallelized operations

GB200 72NVL 72GPU

Models Capabilities (Good or Good enough)

- Language
  - 10-50T Tokens pre-training (10T+ tokens open source now)
  - 100-500B parameters open source (Llama 3 405B)
  - 2-5T parameters closed source (GPT4 is 2T MoE)
- Speech and Voice
  - End to end models, emotion
- Music
  - Not a technology problem but a copy right issue

Source: Mu Li, Boson AI

Models (Good or Good enough)

- Images
  - Resolution improvement beyond 1MP generation
  - Generate images with ‘soul’
- Video
  - SORA, Imagen, etc
  - Expensive to prepare data and train
- Multimodal
  - The trend to combine multiple input/output modalities

Source: Mu Li, Boson AI

State of the art

[Huggingface LMArena]

[Artificial Analysis]

Beyond data, hardware, model for LLM?

- Larger context window
  - Google Gemini 1.5 (2 million token)
- Lower training cost
  - DeepSeek (Technical Report)
    - $~5.5 million, compared to GPT-5 $500 million / 6month
- Lower inference cost
  - Groq, Sambanova, Cerebras (SRAM)
  - 2000-3000 tokens per second

Arena Overview

Scroll to the right to see full stats of each model

First Place | Second Place | Third Place | Default | Compact View

Model 234 / 234	Overall	Hard Prompts	Coding	Math	Creative Writing
claude-opus-4-1-…	1	1	1	1	1
gemini-2.5-pro	1	2	3	1	1
gpt-5-high	1	2	3	1	4
o3-2025-04-16	2	4	3	1	7
chatgpt-4o-lates…	3	5	3	12	2
claude-opus-4-1-…	3	1	1	1	1
gpt-4.5-preview-…	3	5	4	7	1
claude-opus-4-20…	8	5	2	5	2
deepseek-r1-0528	8	8	5	8	7

Semester outline
(Rational Intelligence from Computation and Data, not LLM)

- AI Foundations
- Search
- Constraint Satisfaction
- General Learning
- Machine Learning
- Logic
- Uncertainty
- Natural Language Models

Class Project Start
- Team forming
- Proposal presentation

Class Project Presentation
Final Exam (Closed-book)

Other topics

- Syllabus
  - Rational Intelligence Agent not just LLM
  - In person
  - Assignments & Projects
  - You can use any AI tools and available packages
  - Final 30% closed book

- No class during one week of Nov 2025 (TBD)
  - Business travel